

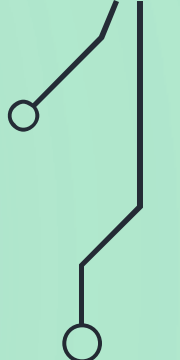
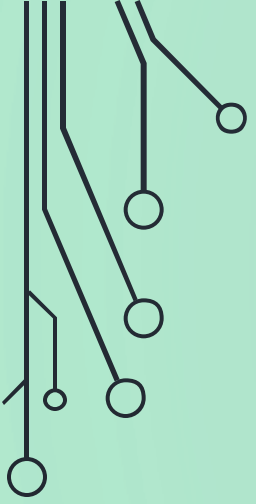
RAG-N-RECIPE

Medically Aware Recipe Generator

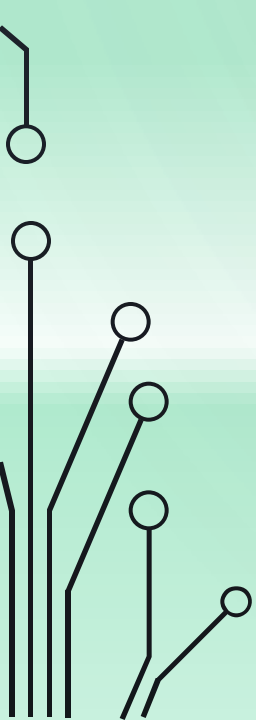
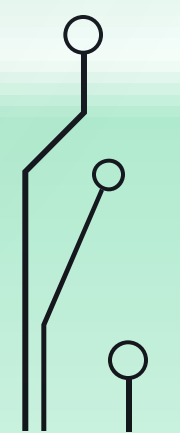
DATS6312_FINAL_PROJECT

ANIRUDH KUMAR RAO





PROBLEM STATEMENT/MOTIVATION

- Many People don't have the luxury of eating just anything due to pre-existing health conditions
 - Generic Recipe Apps don't consider medical constraints
 - Premium Recipe Services are very Expensive \$\$\$
 - Some Powerful LLMS are now free but suffer due to hallucination
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SOLUTION OVERVIEW

- A well established pre trained model (LLM) with a Dual Rag Pipeline
- Side by Side comparison of RAG + LLM vs LLM

Dual RAG Generation Modes



RAG + LLM

- Retrieval-augmented generation
- Indexed ingredient facts
- Medically-aware recipes



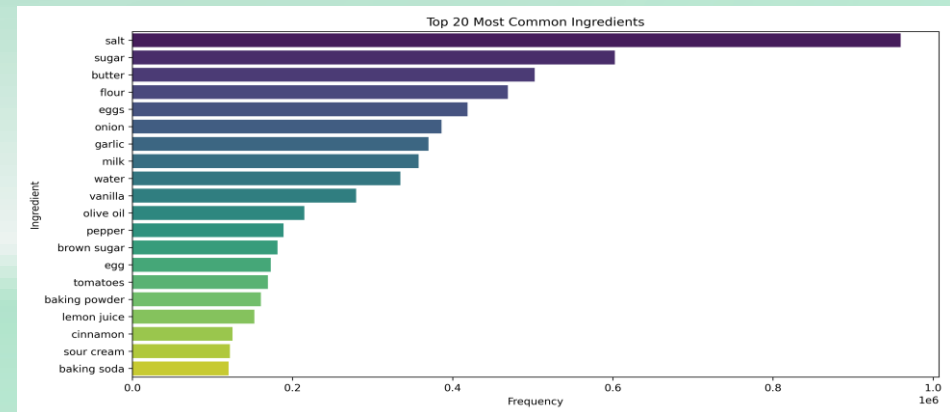
LLM Only

- Purely LLM-based generation
- Pre-trained knowledge
- General recipe suggestions

DATA SOURCES & DATABASES

- Recipes Dataset – Kaggle (2+ Million recipes) – yikes!!
- Ingredients Facts Data – Manually Curated + GPT generated Facts for 50 of the most common ingredients

```
def Label_Data_Rule_Based(row):  
    try:  
        ingredients = eval(row['NER'].lower())  
    except Exception:  
        ingredients = []  
  
    ingredient_text = " ".join(ingredients)  
  
    labels = {  
        "diabetes_safe": True,  
        "hypertension_safe": True,  
        "heart_disease_safe": True,  
        "pregnancy_safe": True,  
        "weight_loss": True,  
        "muscle_gain": True,  
        "gluten_free_safe": True,  
        "lactose_free_safe": True,  
        "kidney_disease_safe": True,  
        "ibs_safe": True,  
        "cholesterol_friendly": True,  
        "pcos_friendly": True,  
        "gout_safe": True,  
        "anemia_support": False,  
        "thyroid_friendly": True,  
        "acid_reflux_safe": True  
    }  
  
    # Diabetes  
    if any(k in ingredient_text for k in ['sugar', 'honey', 'corn syrup', 'sweetened', 'molasses', 'rice', 'noodles']):  
        labels["diabetes_safe"] = False  
  
    # Hypertension & Heart disease  
    if any(k in ingredient_text for k in ['salt', 'soy sauce', 'bacon', 'processed meat', 'sausage']):  
        labels["hypertension_safe"] = False  
        labels["heart_disease_safe"] = False
```



EMBEDDINGS

Pinecone / Anirudh's Org / AI Agents / Database

Docs Settings Feedback Get help AR

Get started Database **Indexes (3)** Backups Assistant Inference API keys Manage

recipies

METRIC	DIMENSIONS	HOST
cosine	384	https://recipies-vx8d5v2.svc.aped-4627-b74a.pinecone.io

CLOUD: AWS REGION: us-east-1 TYPE: Dense CAPACITY MODE: Serverless RECORD COUNT: 20,000

BROWSER METRICS NAMESPACES (1) CONFIGURATION

Records Search List/Fetch Add a record

Namespace: (Default) Search by: ID Top K: 10 Search

+ Filter + Rerank

Showing 10 hits

1 ID: recipe_10
ingredients: ["sweet onion", "champagne vinegar", "poppy seeds", "jumbo lump crabmeat", "fresh grapefruit segments", "canola oil", "avocado", "white grapefruit juice", "watercress", "cherry tomatoes", "...]
instructions: ["Process first five ingredients in a blender or food processor until smooth. While processing, gradually add oil in a steady stream until dressing thickens. Pour into a serving bowl; stir in popp...]
labels: ["pregnancy_safe", "weight_loss", "gluten_free_safe", "lactose_free_safe", "cholesterol_friendly", "gout_safe", "thyroid_friendly"]
title: "Crab, Grapefruit And Watercress Salad"

2 ID: recipe_5499

Indexes Create Index

Search indexes Sort by last viewed Filter by

Showing 3 indexes

recipies Connect

Host: https://recipies-vx8d5v2.svc.aped-4627-b74a.pinecone.io

Cloud: AWS Region: us-east-1 Type: Dense Capacity mode: Serverless Dimension: 384

~~medical_recipies~~ Connect

Host: https://medical_recipies2-vx8d5v2.svc.aped-4627-b74a.pinecone.io

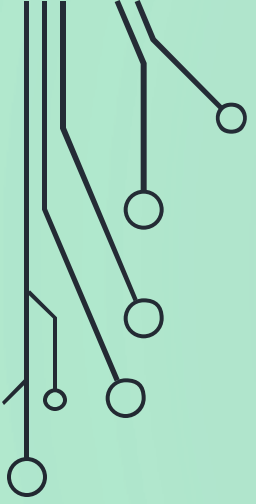
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ingredients Connect

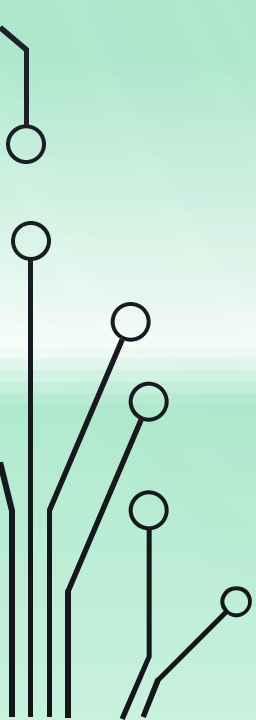
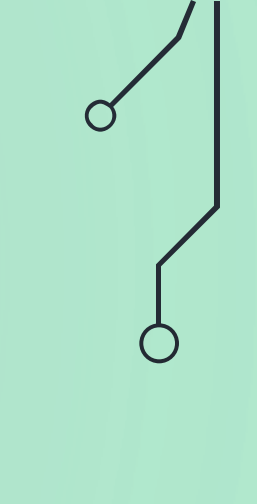
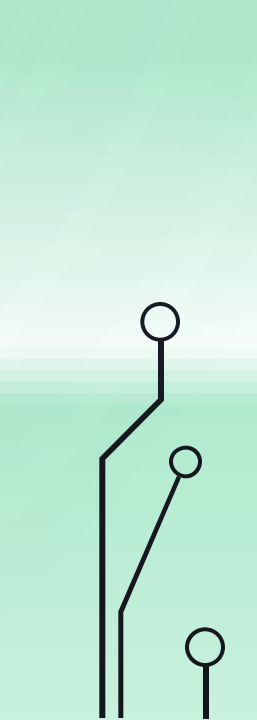
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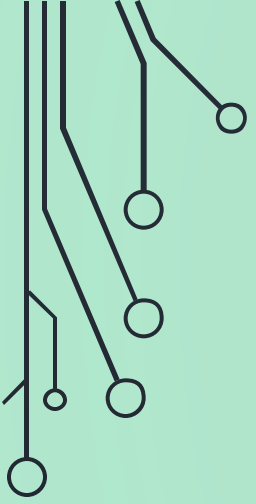
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Embeddings - all-MiniLM-L6-v2 – Sentence Transformers

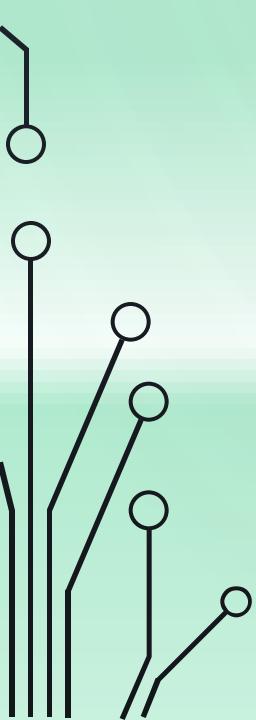


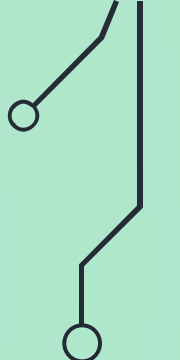
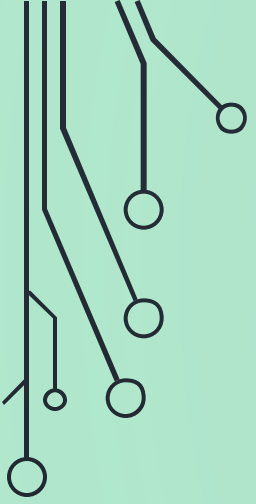
MODEL : MISTRAL-7B-INSTRUCT

- Developed by Mistral AI
 - Causal Language Model – AR Modeling to predict next token
 - Token Context : 8192
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- Max length :6500 ---- Max new tokens :1500
 - Temperature: 0.3
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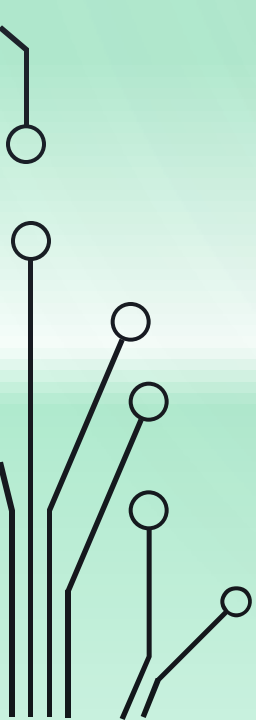


DUAL RAG ARCHITECTURE

- **RAG 1:** Retrieves top-3 recipe matches
 - **RAG 2:** For each ingredient, retrieves top-3 medical facts
 - Prompt constructed using:
 - User Query
 - Recipe Name
 - Ingredients
 - Cooking Instructions
 - Ingredient-level Medical Facts
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LIMITATIONS AND FUTURE WORK

- Fact quality depends on retrieval dataset
 - Finetuning with QLoRA – Was a part of initial proposal but was unable to execute
 - Currently no user personalization or profile memory
 - Future directions:
 - Add patient profiles
 - Evaluate with healthcare experts
 - Nutrition label generation
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